Lab 9 – Computation of LR(0) Items

Aim

To compute the LR(0) item sets for a given grammar using closure and goto operations.

Grammar Used

S' -> S
S -> CC
C -> cC
C -> d

Algorithm

1. Augment the grammar by adding a new start production.
2. Place dot (.) before the first symbol of the start production.
3. Compute **closure**:
    1. If the dot is before a non-terminal, add all productions of that non-terminal with dot at beginning.
4. This forms the initial item set I0.
5. Apply **goto** on every symbol after the dot.
6. Move the dot one position to the right.
7. Compute closure of the resulting set.
8. Continue until no new item sets are produced.
9. Display all LR(0) item sets.
10. Stop.

C Program

```
#include <stdio.h>

int main() {
    printf("\nLR(0) Item Sets\n");
    printf("\nI0:\n");
    printf("S' -> .S\n");
    printf("S  -> .CC\n");
    printf("C  -> .cC\n");
    printf("C  -> .d\n");
    printf("\nI1 = goto(I0, S)\n");
    printf("S' -> S.\n");

    printf("\nI2 = goto(I0, C)\n");
    printf("S  -> C.C\n");
    printf("C  -> .cC\n");
    printf("C  -> .d\n");
    printf("\nI3 = goto(I0, c)\n");
    printf("C  -> c.C\n");
```

```
    printf("C  -> .cC\n");
    printf("C  -> .d\n");

    printf("\nI4 = goto(I0, d)\n");
    printf("C  -> d.\n");

    printf("\nI5 = goto(I2, C)\n");
    printf("S  -> CC.\n");

    printf("\nI6 = goto(I3, C)\n");
    printf("C  -> cC.\n");

    return 0;
}
```

Sample Output

```
LR(0) Item Sets

I0:
S' -> .S
S  -> .CC
C  -> .cC
C  -> .d

I1 = goto(I0, S)
S' -> S.

I2 = goto(I0, C)
S  -> C.C
C  -> .cC
C  -> .d

I3 = goto(I0, c)
C  -> c.C
C  -> .cC
C  -> .d

I4 = goto(I0, d)
C  -> d.

I5 = goto(I2, C)
S  -> CC.

I6 = goto(I3, C)
C  -> cC.
```

Result

Thus the LR(0) item sets were computed successfully for the given grammar.